Understanding the water data gap and emerging technologies to address it: a mini-review

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ABSTRACT

Because water data is critical to many vastly different applications, the problem of developing robust water monitoring infrastructure is often discussed solely in the context of these applications or relative to a few specific indicators rather than as an overarching engineering problem in and of itself. To facilitate an effective synthesis of the application-independent challenges to closing water data gaps at all levels of measurement, we provide a framework for understanding the limitations of current measurement techniques and present a set of 3 engineering goals critical to building a robust water monitoring infrastructure that can meet the demand for higher volumes of data at increased spatial and temporal resolutions. We place special emphasis on the need for development of better instrumentation for in-situ measurement to enable fully autonomous and continuous data collection, and discuss optical methods as the most promising and underutilised class of measurement techniques for accomplishing this.

Keywords: water monitoring, water quality, remote sensing, optical sensing, raman spectroscopy

1 INTRODUCTION

The motivation for this work is rooted in a single question: what's in our water? Our ability to answer this question is crucial for so many aspects of the climate crisis, whether that's ensuring access to clean water or monitoring the health of our oceans or building a global food system able to sustainably feed a growing population in a warming world. And yet, a universal challenge in seeking sustainable alternatives to unsustainable systems is sparsity of data: from carbon pricing to precision agriculture, robust infrastructures for data collection and analysis are in many cases currently a limiting factor to developing the industries at the forefront of the net zero economy. Simply put, we can hardly fix what we can't even measure.

With respect to water monitoring, this data sparsity is particularly egregious. Both the number of observational networks for measurement of water quality and the volume of research into improving water data collection have actually been in steady decline since the mid 1970s (National Resource Council, 2004)(Fekete et al., 2015), despite the fact that the water scarcity crisis is the #3 risk we face globally in terms of impact (Harry Lins, 2018). Currently, approximately 25% of the world's population does not have access to a satisfactory quantity of potable water (Venkataraman, 2023). And the already limited sources from which we obtain freshwater are under further strain from anthropogenic activities and the

environmental changes induced by climate change, with the proliferation of toxic blue-green algae, extreme turbidity, and accelerated eutrophication greatly impacting the sustainability of critical inland water bodies Adjovu et al. (2023).

2 CURRENT IMPACTS OF THE WATER DATA GAP

The implications of water data sparsity span a wide variety of application spaces and interest groups - from industry titans to those whose job it is to regulate them, from scientists to the public, and everyone in between. In this section we present a brief and non-exhaustive overview of three of the most salient categories of problems rooted in insufficient water monitoring resources.

2.1 Industry

Perhaps nowhere is the water data gap more dire than in the highly water-intensive agriculture industry, which consumes roughly 70% of the global water supply. Of that 70%, roughly 40% is lost to the environment (Ritchie, 2021). Additionally via Ritchie (2021), nearly two thirds of nitrogen applied to crops is wasted as runoff which in turn contaminates the surrounding environment. This resource waste is massive due to a longstanding practice of overfertilizing and overwatering because precise resource demands of crops are not known. In the agricultural product chain, imprecise resource allocation doesn't just result in wasted inputs, but also diminished outputs. According to Dzombak (2021), growing evidence suggests that contrary to longstanding belief, excess fertilization is resulting in lower crop yields during droughts by creating eutrophic conditions in which plants are starved of oxygen.

2.2 Public Health

Sparsity of data with respect to drinking water and wastewater poses a major public health concern, and further invisibilizes the injustices faced by frontline communities facing the most risk from industrial pollution and climate volatility. The existing infrastructure for monitoring public water sources in the US has a history of producing insufficient results at best, and blatantly inaccurate results at worst. An estimated 26-38% of health-related violations are either not reported or inaccurately reported to the national Safe Drinking Water Information System according to Josset et al. (2019), whose analyses further find that 28% of dissolved oxygen data in the US Water Quality Portal exhibits inconsistent or even physically unfeasible values. Additionally, there are critical absences of standardized methods for measuring a number of potentially extremely harmful substances in water, particularly "forever chemicals" such as PFAS. In a 2018 report on addressing these data gaps relating to PFAS in drinking water, the EPA cites numerous superfund sites across the U.S. that are currently contaminated with "high concentrations of tentatively identified compounds having little or no toxicity information" (EPA, 2018).

2.3 Environment

Long term environmental water quality monitoring programs are of vital importance to mitigating the impacts of climate change, informing public policy, and assessing the success of remediation programs. Inherent in the irrelevance of borders to global ecosystems is the need for high-level coordination between governments which is extremely lacking; in the U.S., state-level agencies are responsible for the monitoring and enforcement of national regulations such as the Clean Water Act, and the role of federal agencies is in large part limited to the aggregation and analysis of state-level datasets (Josset et al., 2019). Effective communication between federal and local governments, indigenous nations, nonprofit organizations, and industry is a ubiquitous barrier to the establishment of effective long-term monitoring infrastructure

(Miller et al., 2022) which magnifies in intensity all the way up to the international level (United Nations Environment Programme, 2023). This leads to major inefficiencies that add to the already extremely high cost of monitoring programs, which require extensive planning and are typically highly labor intensive, involving numerous field crews and travel to individual, carefully chosen survey sites numbering in the hundreds or even thousands (A. Kiddon et al., 2020).

As a result of this lack of coordination and high-level planning, serious limitations to data accessibility and reusability often arise because data obtained from a particular study tend to be tailored to a few specific points of inquiry and lack the spatial or temporal resolution needed to investigate questions posed by other researchers (Miller et al., 2022). The lack of continuous water flow and water quality monitoring data is a serious impediment to environmentally sound and well-informed policymaking, appearing as a persistent problem in a whole host of applications from decisions about groundwater withdrawals (National Resource Council, 2004) to preventing eutrophication of estuaries (Fertig et al., 2014). The high cost and logistical capacity required to conduct water quality surveys means that oftentimes data needed for certain areas are absent entirely, leaving important questions like the extent to which shale gas development contaminates groundwater (Betanzo, 2016) or the rate of plastic accumulation in earth's oceans (United Nations Environment Programme, 2023) unanswered.

3 WATER DATA TYPES

Water quality is generally defined relative to some subset of physical, chemical, and biological characteristics that are relevant to the water source under study in a particular application. A subset of common parameters and methods used to measure them is aggregated in Table 1. Hydrological, limnological, and oceanographic studies typically use "indicators", or combinations of water quality parameters that are statistically combined, in order to to quantify some qualitative characteristic to evaluate water quality on an application-relevant, standardized basis (Fertig et al., 2014). For example, Fertig et al. (2014) use a combination of 15 parameters (including temperature, dissolved oxygen, chlorophyll α, and total nitrogen concentration, among others) to define 3 indicators - water quality, light availability, and seagrass response - which they use as an index to quantify eutrophication. Alternatively, Ochungo et al. (2019) assess potability of groundwater by synthesizing 8 parameters (including electrical conductivity, sulfate, potassium, sodium, and other aqueous ions) into a single index that allows them to compare samples on a letter grading scale from A ("excellent") to F ("undrinkable"). While the terms "parameter" and indicator" are often used interchangeably in practice, within this work we use "parameter" in reference to the individual analytes aggregated in Table 1 and "indicator" to refer to application-specific metrics created via statistical synthesis of some combination of parameters.

4 WATER MEASUREMENT TOOLS

Techniques for measuring water quality can generally be classified as either remote or in-situ, and these two groups of techniques serve complementary purposes. In-situ measurements involve direct sampling, either manually or by an autonomous system deployed in the field, whereas remote sensing typically refers to a relatively newer class of indirect measurements derived from airborne or satellite observations. When used in tandem, remote and in-situ observations can together provide a much more complete data picture than either approach alone - for example, in-situ measurements can be used to measure exact concentrations of pollutants and rivers and sea waters, while remote sensing measurements can be used to trace propagation of pollution over time (Myroshnychenko et al., 2015).

Remote sensing (RS) techniques use solar reflectance in a variety of frequency bands along the electromagnetic spectrum to gather data on water quality parameters. Optical RS techniques are the oldest and most widely used, and are mainly passive; their spectral imaging measurements are based purely on reflected solar energy. Microwave RS techniques are newer and exist in both the passive and active forms, with some sensors such as RADAR and LIDAR sending pulses of radiation in the microwave band to deduce target properties with less unwanted interference from the atmosphere (Cho et al., 2022). The two main physical water parameters that affect the energy levels detected by RS techniques are water color and turbidity (Adjovu et al., 2023). Water color increases the absorption of light in water, thereby reducing the energy flux reaching the sensor, whereas increased turbidity increases the energy flux reaching the sensor because more solar energy is backscattered by higher particle densities. In general, remote sensing is useful for chemical and biological analytes which are spectrally active in the visible and IR portions of the electromagnetic spectrum (Adjovu et al., 2023). Such parameters include chlorophyll-α, algae, total dissolved organic matter, dissolved organic carbon, and pH. Weaker parameters including nitrogen, dissolved phosphorus, dissolved oxygen, and micro-pollutants can be estimated via inference with remote sensing data but come with much greater uncertainties.

One useful way of classifying in-situ methods is by whether they require collection and alteration of a sample, or can be performed continuously and non-destructively in an immersed environment. Examples of the former category include evaporation, filtration, titration, and most varieties of colorimetry (Alley, 2007). The latter category includes electrometry and optical techniques such as UV absorption spectroscopy, IR fluorescence spectroscopy, and raman spectroscopy (Brian Pellerin et al., 2012). Although they incur a much larger logistical and monetary cost, the in-situ measurements done by collecting samples and altering them under controlled lab environments are important tools to use as the ultimate ”ground truth” for informing calibrations of other faster and more scalable methods that rely on inference and statistical associations.

While certain types of in-situ measurements can be considered indirect (e.g. optical methods which use calibration to convert spectral peak intensities to chemical concentrations), it’s important to note that *all* remotely sensed data require calibration against directly measurements, and therefore remote sensing can never fully replace in-situ measurements despite it’s potential to greatly augment our water monitoring capabilities beyond what could be achieved using in-situ monitoring alone (Fekete et al., 2015). Because remotely sensed data carry additional time-varying biases due to atmospheric and land surface interference, calibration and analysis of these datasets require an additional layer of complexity (National Resource Council, 2004). However, another important consideration is that the irrelevance of borders and surface barriers to satellite measurements makes fully automated, continuous monitoring and accessible sharing of data far easier to achieve with remote sensing than with in-situ measurement (United Nations Environment Programme, 2023).

5 SOLUTION FRAMEWORK

Despite the highly positive (approximately 4 to 1) benefit to cost ratio of investment in public water data infrastructure (Albert Cho et al., 2017), decades of insufficient funding has led to a steady decline in data coverage. This deficit is exacerbated by the global volatility imposed by climate change and the increasing demand for data in tandem with global water consumption. Such limited resources in the presence of such urgent necessity represent an imperative for innovation towards water monitoring techniques that can produce more data not only at lower costs, but also with higher accuracy and with less human time and labor.

We identify three key components to closing the water data gap, where substantial progress is essential in all three to build a functional and effective global water data repository. We discuss each in further detail in the following sections[1].

5.1 Improving data accessibility and standardization

The first point is where progress can most cheaply be made, but it presents substantial political and logistical challenges. While there is indeed a sparsity of data due to limitations imposed by our capacity for observation (see sections 5.2 and 5.3), there is also a substantial gap between the data that exists and the data that can be accessed by those who wish to use it. Fekete et al. (2015) succinctly pinpoint the basis for this intractable problem of data accessibility, remarking that ”data sharing is voluntary, and the (often state-affiliated) agencies collecting observations rarely have the obligation or incentives to share”. Additionally, the use of data that is accessible is often further hindered by a lack of *standardization*, or consistency in measurement methods. Wide variances in spatial and temporal resolution as well as observed analytes are often cited as a major limitation in usability of current water quality data (Fertig et al., 2014)(Miller et al., 2022). These problems point to a strong need for legislation and international cooperation to endorse data sharing, develop standard water monitoring frameworks, and implement cooperative long-term water monitoring programs.

5.2 Expanding capacity for remote monitoring

Increasing investment in airborne and spaceborne remote sensing programs is an immensely cost-effective way to increase the volume of accessible data for a variety of water quality parameters - particularly those which are optically active as discussed in section 4. There is already substantial empirical evidence in support of this; the utilization of Landsat image data for water quality assessment has saved global users approximately 3.45 billion USD as of 2017 since the inception of the program in the 1970s (Anna E. Normand, 2021). However, the extent to which high spatial resolution is prioritized imposes a cost tradeoff; while airborne sensors have finer spatial and temporal resolution than spaceborne sensors, the data often come at a substantially higher price point (Adjovu et al., 2023), although increasing accessibility of UAVs could soon alter this paradigm. One significant opportunity for expanded remote sensing capability is the development of microwave sensors, which are less susceptible to atmospheric attenuations and are capable of penetrating further depths of surface water (Khanal et al., 2020). Additionally, emerging computational methods such as neural networks and novel atmospheric correction algorithms show significant promise for improving the accuracy of water quality parameter models based on remote sensing data (Adjovu et al., 2023). However, the most effective improvement in this respect will likely come from increased volume of in-situ measurements for data calibration and validation. (Dey and Vijay, 2021)(Gholizadeh et al., 2016)(Fekete et al., 2015)

5.3 Expanding capacity for in-situ monitoring

And therefore, expanding capacity for in-situ observation is arguably the challenge that is both the most complex *and* of the most urgent importance, because it is impossible to completely replace in-situ monitoring with remote sensing despite the vastly greater logistical ease with which accessible, spatially and temporally resolved data can be obtained via satellites. Extensive developments in the engineering of instrumentation and data architectures will be required to bring this to fruition.

[1] We use the key phrase ”expanding capacity” to mean both increasing the volume of data collected and improving upon the spatial and temporal resolution of that data

The key weaknesses of in-situ measurement can be summarized as follows:

- expensive
- labor-intensive
- time-consuming
- not suitable for large-scale analysis

Each measurement technique discussed in section 4 exhibits some or all of these characteristics, which begs the question of how exactly these limitations can be transformed into engineering goals. One readily apparent approach is to work towards replacing lab measurement techniques with tools that can be operated fully in the field, ideally with little to no direct human interaction (Brian Pellerin et al., 2012). Currently, the vast majority of in-situ monitoring programs are still highly manual in nature, run by teams of researchers who spend countless hours planning survey locations, collecting samples in the field, and analyzing those samples in labs in order to obtain results (A. Kiddon et al., 2020). For governments and NGOs struggling to allocate limited resources, it's quite reasonable to conclude that scaling up these monitoring programs as they exist now won't make enough of a dent in the data gap to be worth the investment - especially as the complexity and scale of demand for new data to inform emerging technologies only continues to increase. If investments were instead made in the engineering of more advanced, rugged instrumentation for autonomously conducting suites of water quality measurements on-site, truly scalable in-situ monitoring would become far more feasible through the deployment of distributed networks of monitoring stations with immersion probes running on automated measurement schedules which provide continuous data streams.

Because they require no chemical alteration of the sample and are capable of measuring a variety of target analytes within the same wavelength range, optical measurement systems are arguably one of the most promising avenues towards realizing truly autonomous, large-scale in-situ monitoring. The optical methods discussed in section 4, however, still present serious limitations that must be addressed before they can be utilized in this manner. One key consideration is that because optical methods are indirect in the sense that they rely on empirical calibration data to convert spectral peak intensities to concentrations, they are vulnerable to similar sources of inaccuracy that must be considered with remote sensing data (Brian Pellerin et al., 2013). Fluctuating environmental conditions cause observed spectral baselines to shift over time, introducing an additional source of uncertainty. Furthermore, the problem of engineering optical systems rugged enough to hold up under sustained exposure to harsh field conditions (e.g. sunlight, saltwater) is nontrivial, and careful design of optical probes and calibration techniques is required to account for a large variety of interferences (e.g. air bubbles, stray light, pH, biofouling, turbidity) that can compromise data entirely (**?**). Existing in-situ optical probes on the market often incorporate automated mechanism for keeping the lens clean which have been found to be effective at dealing with interferences (Saraceno et al., 2009), but require accounting for higher design complexity and additional potential modes of failure.

Raman spectroscopy is perhaps the most underutilized among the existing optical methods for water measurement due to the high sensitivity required to detect the extremely weak raman scattering phenomenon. Despite this, raman spectroscopy is uniquely well suited to water measurement over other optical methods in that it is particularly sensitive to protein structure and organic compounds, it can be used on fresh or saltwater samples, and water has a much lower background signal in raman spectra than in UV or NIR spectra. Investing in the development of low cost, portable raman systems for use in field applications therefore holds great potential to revolutionize water monitoring. One example of progress currently being

made in this realm is in the work of Atabaki et al. (2021) and Persits et al. (2021) [2], who are working to build ”swept-source” raman systems which are able to operate using lower power lasers yet achieve much higher signal to noise ratios than standard benchtop spectrometers by using tunable source lasers to replicate the dispersive mechanism of a spectrometer without the limitation on optical throughput to the system imposed by the spectrometer slit.

6 CONCLUSIONS AND OUTLOOK

The strategies for mitigating the adverse effects of climate change that we choose to implement in this decade will have dramatic repercussions, for better or for worse, for centuries to come. It is absolutely vital that we seize this opportunity to not only adapt to increasing global volatility, but also develop technology and policy that facilitates evolution towards a more just, equitable, and sustainable global socioeconomic order. While a significant progress is being made in this direction, it is important to also consider that success cannot be achieved without the capacity to correctly identify the problems at hand *and* assess the effectiveness of proposed solutions. Although the urgency of the crises we face makes it tempting to prioritize investment in technology that will address symptoms over technology that will diagnose the root problems, we cannot afford to continue neglecting the underdeveloped and underfunded data collection architectures that hold the key to understanding the effects of our actions on the world around us.

The water data gap is arguably the most egregious example of the limitations to progress imposed by this pattern of data neglect, and the task of closing it demands collaboration and coordination on a global scale to an extent far beyond what has been achieved to date. Beyond just improving the accessibility of current data, it is equally important that substantial resources and effort are invested in dramatically expanding the scope and volume of long-term monitoring programs while also leveraging emerging technologies to transition heavily towards autonomous data collection both via in-situ and remote sensing methods. Autonomous, durable, and scalable monitoring infrastructures are the bedrock upon which we will develop and deploy data informed solutions to build a sustainable future, and it is of the utmost importance that we invest the proportionate resources now to make that foundation rock solid.

[2] NOTE: this is the topic of the author’s current research

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TABLES

Parameter	Units	Measurement Techniques/Tools	Notes
Turbidity	TU, NTU	nephelometric turbidimeter, remote sensing	1 TU=1 mg/L of silica in suspension
Temperature	°C, °F	thermometer/thermistor, remote sensing w / raman spectroscopy [3]	
Color	PCU	visual (color standards), spectrophotometry, remote sensing	PCU are derived from Pt-Co standard, 0-500 range
Total Suspended Solids (TSS)	mg/L	filtration	Total Solids (TS)= TSS+TDS
Total Dissolved Solids (TDS)	mg/L	evaporation	water can be classified by the amount of TDS per liter as follows - freshwater: ¡1500 mg/L, brackish: 1500-5000 mg/L, saline water: ¿5000 mg/L
Electrical Conductivity	mS/m, dS/m	electrometric method	Electrical conductivity and TDS are related and can be used to estimate one another. One of the main parameters used to determine suitability of water for irrigation and firefighting.

(a)

Table 1. Overview of common physical (a), chemical (b), and biological (c) water quality parameters along with the techniques used to measure them and typical uses. Information from Hassan Omer (2020) and Alley (2007) unless otherwise noted.

[3] Artlett and Pask (2017)

[4] Tabatabai (1974)

[5] Brian Pellerin et al. (2013)

[6] G/Mariam et al. (2021)

Parameter	Units	Measurement Techniques/Tools	Notes
pH	logarithmic	electrometry, colorimetry	highly acidic water allows heavy metals to dissolve more easily; ammonia is relatively harmless in neutral or acidic water, but becomes progressively more poisonous as pH increases
Chloride	mg/L	titration by silver nitrate	>250 mg/L chloride may indicate wastewater pollution
Sulfate	mg/L	turbidimetry [4], colorimetry	Typically present due to leaching of sodium sulfate or magnesium sulfate deposits
Nitrogen	mg/L	UV absorption spectroscopy, colorimetry, electrometry [5]	There are four forms of nitrogen in water and wastewater: organic nitrogen, ammonia, nitrite, and nitrate.
Fluoride	mg/L	electrometry, UV-Vis spectrophotometry [6]	Around 1 mg/L of fluoride is effective in preventing tooth decay; the maximum allowed quantity varies between 1.4 mg/L and 2.4 mg/L
Dissolved Oxygen	mg/L	colorimetry, Winkler titration, electrometry	one of the key parameters used to evaluate water quality of streams, rivers, and lakes; higher dissolved oxygen is indicative of better water quality
Toxic Inorganic Substances	mg/L	atomic absorption photometry (metals), inductively coupled plasma (metals), spectrophotometry, colorimetry, electrometry, titration	includes metals (e.g. lead, mercury, arsenic), and nonmetals (e.g. nitrates, cyanides)
Toxic Organic Substances	mg/L	gas chromatography, high performance liquid chromatography, mass spectrophotometry	usually anthropogenic pollutants - pesticides, detergents, etc
Radioactive Substances	pCi/L	gross alpha method, gross beta method	similar to TDS method

(b)

Parameter	Units	Measurement Techniques/Tools	Notes
Biochemical Oxygen Demand	mg/L	winkler titration, electrometry	often used as a measure of the power of sewage, key indicator of eutrophication
Bacteria	CFU/L	plate count method	CFU = colony forming unit. Cause of many dangerous waterborne diseases (cholera, paratyphoid fever, etc)
Viruses		PCR	Waterborne viruses are known to cause infectious hepatitis and poliomyelitis. Most can be deactivated by disinfection at water treatment plants.
Chlorophyll α	mg/L	fluorescence, spectrophotometry, evaporation	Measure of algae concentration in water. Certain species of algae (e.g. blue-green algae) are toxic. Indicator of eutrophication - correlated to high nitrate and phosphate levels.

(c)